

Junnosuke Kamohara

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Education

MS in Aerospace Engineering, Tohoku University | Sendai Japan

Oct 2023-the present

Graduate School of Engineering, Department of Aerospace Engineering

BS in Aerospace Engineering, Tohoku University | Sendai Japan

Apr 2020-Sep 2023

Faculty of Mechanical and Aerospace Engineering | 1 semester early graduation, GPA: 3.53/4.00

Robotics Projects

Revival project of a moonraker, a flight model from Google LunarXprize

Oct 2023 - the present

Space Robotics Laboratory, Personal project

- Update the rover from the past project and make it compatible with **ROS2**.
- Redesign the electronics system and create PCB boards.
- Controlled Maxon EC-max motor by **micro-ROS-Arduino**.
- Created ROS2 package for PS4 teleoperation by implementing, in **C++**, a four-wheel controller using a differential drive.

Development of a photorealistic lunar simulator and Sim-to-Real of rock segmentation.

June 2022 - Sep 2023

Space Robotics Laboratory, Undergraduate Research

- Developed an **open-source photorealistic lunar simulator** and evaluated its photorealism against real-world data.
- Developed a lunar simulator based on **Omniverse Isaac-Sim** with Dr. Antoine Richard from SpaceR, the University of Luxembourg.
- Worked on the sim-to-real of **YoloV8** instance segmentation model in the lunar analog environment. Synthetic data with automatic annotation are generated with **Omniverse Replicator**, and actual data recorded by RealSense D435 are manually labeled with RoboFlow.

Behavior planning in the office environment

Sep 2023 - the present

Guide Robotics Inc., Internship project supervised by Dr. Aweek Das

- Develop the behavior planning system based on ROS2 navigation (**Nav2**).
- Integrate Nav2 with our own closed-sourced **VIO** algorithm.
- Developed the PoC behavior planning system with Clearpath **Turtlebot4**.

Experiments on Luxonis OakD camera for SLAM and hardware inference of NN

July 2023 - Sep 2023

Guide Robotics Inc., Internship project supervised by Dr. Aweek Das

- Created a report of **DepthAI ROS** for stereo camera and IMU sensing.
- Worked on **OpenCV chessboard calibration** of a stereo camera.
- Implemented the program to convert the **Darknet yolov4-tiny** model into **MyriadX blob** format to run neural networks on the camera's RVC2 chip.

Benchmark of open-source visual-inertial SLAM algorithms and our own close-sourced system

Apr 2023 - June 2023

Guide Robotics Inc., Internship project supervised by Dr. Aweek Das

- Compared the performance of open-source visual SLAM: **ORB-SLAM3** and **Elbrus-SLAM** and our close-sourced **VIO** algorithm.
- Compared the estimated trajectory and ground truth from RTK GPS and calculated the error by **evo**, a Python package.
- Verified that our VIO realizes **much less drift** than VIO algorithms.

Data-driven simulator for autonomous driving

June 2022 - Dec 2023(expected)

Joint research with Japanese autonomous driving startup TIER IV.inc.

- **Goal:** Develop a data-driven simulator to enable photorealistic camera and LiDAR simulation.
- Image and LiDAR point cloud synthesis and surface extraction using **NeRF** and **NeuS** and the custom dataset recorded by TIER IV. Revealed **SDF-based model** produces much better sensory data and surface than NeRF-based model regarding geometric consistency.
- Implemented **Urban Radiance Fields** from Google Research. However, the NeRF-based solution suffered poor geometry reconstruction in the outdoor environment with a few multi-view images.
- Implemented **Python dataset class** and **PyTorch dataloader** to convert the dataset in **MMDetection3D** format into the data format compatible with NeRF.
- Evaluated the unofficial implementation of **Block-NeRF**. Tested the model performance in **Waymo Block-NeRF dataset** and custom dataset from TIER IV.

Unsupervised domain adaptation in LiDAR semantic segmentation

Nov 2021 - Mar 2023

Joint research with Japanese autonomous driving startup TIER IV.inc.

- Implemented **unsupervised domain adaptation** algorithm from this paper(<https://arxiv.org/abs/2107.09783>).
- Evaluated the effect of domain adaptation on LiDAR semantic segmentation model called **SalsaNext**.
- Unfortunately, the proposed algorithm did not fill the domain gap between **Velodyne HDL64** LiDAR, the sensor used for training the pre-trained model, and **VLS128** LiDAR, the sensor used in TIER IV data.

Publication

[Conference Papers]

J. Kamohara, K. Sakamoto, A. Ohsato, S. Tomie, M. Kataoka, M. Kawano, "Development and evaluation of Neural Simulator for autonomous driving systems," The 38th Annual Conference of the Japanese Society for Artificial Intelligence, 2024.

A. Richard*, **J. Kamohara***, K. Uno, S. Shreya, D. van der Meer, M. Olivares-Mendez, K. Yoshida, "OmniLRS: A Photorealistic Simulator for Lunar Robotics," International Conference on Robotics and Automation, 2024. (*equal contribution)

K. Tabata, **J. Kamohara**, R. Unno, M. Sato, T. Watanabe, T. Kume, Y. Negishi, R. Okada, Y. Iwasawa, Y. Matsuo, "Construction and Validation of Action-Conditioned VideoGPT", The 37th Annual Conference of the Japanese Society for Artificial Intelligence, 2023.

Y. Negishi, M. Sato, R. Unno, K. Tabata, T. Watanabe, **J. Kamohara**, T. Kume, R. Okada, Y. Iwasawa, Y. Matsuo, "Scaling Laws of Dataset Size for VideoGPT", The 37th Annual Conference of the Japanese Society for Artificial Intelligence, 2023.

M. Sato, R. Unno, Y. Negishi, K. Tabata, T. Watanabe, **J. Kamohara**, T. Kume, R. Okada, Y. Iwasawa, Y. Matsuo, "Scaling Laws of Model Size for World Models", The 37th Annual Conference of the Japanese Society for Artificial Intelligence, 2023.

[Presentation]

J. Kamohara, K. Sakamoto, A. Ohsato, S. Tomie, M. Kataoka, M. Kawano, "Development and evaluation of Neural Simulator for autonomous driving systems," The 38th Annual Conference of the Japanese Society for Artificial Intelligence, Hamamatsu, Japan, May, 2024.

A. Richard*, **J. Kamohara***, K. Uno, S. Shreya, D. van der Meer, M. Olivares-Mendez, K. Yoshida, "OmniLRS: A Photorealistic Simulator for Lunar Robotics," International Conference on Robotics and Automation, Yokohama, Japan, May, 2024.

A. Richard, D. van der Meer, G. Garcia, L. Chovet, **J. Kamohara**, S. Bogh, "Towards a common robotics simulator for lunar environments?," IEEE SMC-IT & SCC 2023, Caltech Pasadena, CA, 2023.

Awards

Nakajima Foundation Scholarship Program

Sep 2024 - Aug 2029

The program supports tuition (maximum 3 million JPY, equivalent to **20,000 USD**) and a stipend of 300,000 JPY, equivalent to **2,000 USD** a month for living expenses. Tuition is supported for the first two years.

Job Experience

Guide Robotics, Robotics & AI Intern | Tokyo, Ginza

Dec 2022 - the present

- Benchmarked various VIO algorithms (close-sourced VIO system, **ORB-SLAM3**, **Elbrus-SLAM**) with outdoor data from clients.
- Tested visual-inertial SLAM on Luxonis OakD camera.
- Worked on the simulator for warehouse and office environments using **Isaac-Sim** and **Gazebo Ignition**.
- Developed **Nav2** based behavior planning in the office environment.

Matsuo Institute, AI Intern | Tokyo, Bunkyo

Nov 2021 - Dec 2023(expected)

- Neural camera and LiDAR simulations using **neural implicit models**.
- Extensive survey on **World Models** in autonomous driving.
- **Unsupervised domain adaptation** of point cloud semantic segmentation.
- Construction of **action-conditioned Video-GPT**, the video prediction model.
- Evaluation of **scaling law** in action-conditioned Video-GPT.

Zero to One, AI Intern | Tokyo, Roppongi

Mar 2021 - Dec 2022

- Developed the chat system to train **medical students** for the interview with patients.
 - Question understanding system
 - * Embed questions by pre-trained **Multilingual Sentence Transformer**.
 - * Predict question label using **K-nearest neighbor** in embedding space.
 - Response system
 - * Query expected answers from the response database.
- Created teaching materials for the Japanese deep learning qualification exam (JDLA Deep Learning for Engineer).
 - Created exercise program.
 - * PyTorch implementation.
 - * TensorFlow2 implementation.
 - Created slides to explain concepts of various neural network architectures and algorithms.
 - * Fundamental neural network architecture (MLP, CNN, RNN).
 - * Generative models (VAE, GANs).
 - * Reinforcement Learning (Q-learning, DQN, A2C, A3C).
 - * Explainable AI (SHAP, Grad-CAM).
 - * Language Models (Seq2Seq, Transformer, BERT, GPT).

Skills

Programming Python, C/C++, MATLAB, Git

Robotics ROS 1/2, Omniverse Isaac-Sim, Gazebo, Arduino, Turtlebot4, 2/4-wheeled rover

Software Pytorch, Tensorflow, Docker, OpenCV, Solidworks, Fusion360, Eagle, Nvidia Omniverse

Hardware PCB design, Milling machine, Lathe, Drill, 3D printer

Interest

In graduate school, I am keen to work on the following research topics:

- Autonomous task allocation in Multi-Robot System.
- Active perception with Multi-Robot System.
- Hierarchical motion planning using geometric and semantic representation such as 3D scene graph.
- Construction of hierarchical semantic and geometric map representation using neural implicit models.
- Safe and robust imitation learning, which is resilient to out-of-the-distribution.